

Multi-Objective Genetic Algorithms in Designing Redundant Water Distribution Systems

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Abstract—Water supply and distribution systems are critical infrastructures whose design heavily influences the operating conditions during unexpected demands. Rapid urbanization, increasing leakage and ageing infrastructures are the main factors responsible for poor performance under different scenarios. Water distribution networks have to be designed with the aim of reliability, but also with the need of containing costs. In these last decades, several Evolutionary Algorithms have been proposed for solving such problems, obtaining successful results. The paper aims at comparing the performance of two multi-objective genetic algorithms in solving a relatively simple optimization problem, for which the complete Pareto front has been calculated. The tests have been performed on the problem of designing the pipes of a two-loop water distribution network where the first objective is the minimization of the cost and the second objective is the maximization of the redundancy, expressed as the average pressure surplus above a minimum required value. The performance indices considered keep into account both the exploration and the exploitation capabilities of the algorithms.

Keywords—Water supply, Pressure, Leakages, Reliability, Pareto frontier

I. INTRODUCTION

Multi-objective evolutionary algorithms (EAs) for the planning and management of water supply systems have been increasingly adopted in these last decades, especially for the resolution of problems with two criteria, the first generally represented by costs and the second expressed as some measure of the network performance.

References [1] and [2] were among the first to adopt, respectively, a single and a multi-objective EA for the rehabilitation of a pressurized network system, minimizing costs and, in the second case, maximizing also the benefits. Besides the minimization of total cost, [3] introduced the maximization of a resilience index, [4] the minimization of the maximum pressure deficit at nodes, while [5] took into consideration the reliability of the network as an objective to be maximized, an approach followed also by [6].

In subsequent years, several other approaches were proposed. [7] introduced a genetically adaptive multi-objective algorithm, [8] a multi-objective particle swarm optimization, [9] a multi-objective cuckoo-search, while [10] a multi-objective genetic algorithm for least cost design and resiliency maximization.

The best-known approximation of the true Pareto front using five standard multi-objective evolutionary algorithms was

obtained by [11], while [12] used the decomposition method to show that it performs better than NSGA-II [13]. A new hybrid multi-objective EA was further developed by [14]. An approach based on non-dominated sorting differential evolution was developed by [15] and [16] for water distribution systems (WDS) optimization, including cost and resilience. In these last years, further research has been carried out by many authors, especially on improving the exploitation capabilities of the algorithms, [17] and [18]; in particular, [19] adopted a hybrid NSGA-II algorithm, augmented with a random multi-point crossover operator.

In this work, the performance of two multi-objective genetic algorithms (GAs) has been analyzed with reference to a two-objective optimization problem, for which the Pareto frontier has been thoroughly calculated. The tests have been performed on the problem of designing the size of the pipes of a two-loop water distribution network for which minimization of cost and maximization of redundancy are the main objectives. A global performance index (the hypervolume, [20]) was compared with some more specific indices already introduced by [21], quantifying the performance of the solutions in terms of exploration and exploitation in approaching the true Pareto frontier.

The paper is organized as follows: Section II presents the mathematical formulation of the problem and the benchmark network, while Section III illustrates the two multi-objective GAs adopted, together with the performance metrics considered; Section IV shows the results obtained, and Section V draws some concluding remarks.

II. THE OPTIMIZATION PROBLEM AND THE BENCHMARK NETWORK

The problem of WDS optimization with two conflicting objectives is usually aimed at maximizing some performance of the network, for example in terms of the reliability, of the resilience, or of the redundancy, and minimizing the total costs for construction and management. In this work, given the theoretical nature of the analyses, a relatively simple problem has been assumed, in order to calculate the entire Pareto front.

A. Analytical formulation

The problem is formulated as composed of two objective functions: the first is aimed at minimizing the overall cost of the pipes in the network; the second is the maximization of the average pressure surplus at the nodes, with reference to a minimum design threshold. In this context, the pressure value at

each node should always be above the required level, which is a constraint for delivering adequate flow to customers. Actually, according to the several scenarios and variable working conditions to which a system is subjected, both negative deviations (deficit) and pressure surplus from the required minimum are possible. The mathematical formulation is:

$$\min f_1(d_1, d_2, \dots, d_{N_p}) = \sum_{i \in D} \sum_{j=1}^{N_p} c(d_i) L_{ij} \quad (1)$$

$$\max f_2(d_1, d_2, \dots, d_{N_p}) = \sum_{k=1}^{N_n} \frac{H_k - H_k^{req}}{N_n} \quad (2)$$

Eq. (1) and eq. (2) are subjected to the following physical constraints:

$$\sum_{i \in n_k} Q_i - \sum_{j \in m_k} Q_j = Q_{e,k} \quad (3)$$

$$\sum_{i \in p_j} h_{f,i} = \Delta E_j \quad (4)$$

Eq. (3) represents water balance (continuity) at each node, while eq. (4) the energy conservation along path j . Moreover, the following constraints have to be satisfied:

$$H_k \geq H_k^{req} \quad (5)$$

$$d_{\min} \leq d_i \leq d_{\max}, \quad i \in \mathbb{N} \quad (6)$$

Eq. (5) requires a minimum required value at each junction, while eq. (6) constrains the problem to a discrete set of commercial diameters. In the preceding equations, the symbols have the following meaning:

- $d_1, d_2, \dots, d_{N_p}$ are the N_p decision variables representing the number of pipes to be designed

- $c(d_i)$ and L_{ij} are the cost per unit length and the total length of the link j with diameter d_i

- D is the set of the N_D available commercial diameters (ranging from a minimum value, $d_{\min}$, and a maximum, $d_{\max}$)

- N_n is the number of nodes where flow delivery to customers takes place (they are also called junctions); each junction, k , is characterized by an actual value of piezometric head, H_k , and a required one, H_k^{req} (note that the piezometric head is the sum of the ground elevation and the pressure)

- $Q_{i,k}$ and $Q_{j,k}$ represent the n_k and m_k flows, respectively, entering or leaving junction k

- $Q_{e,k}$ is the delivered flow to customers at junction k

- p_j is the number of links belonging to path j

- $h_{f,i}$ is the head (energy) loss along link i of the path j

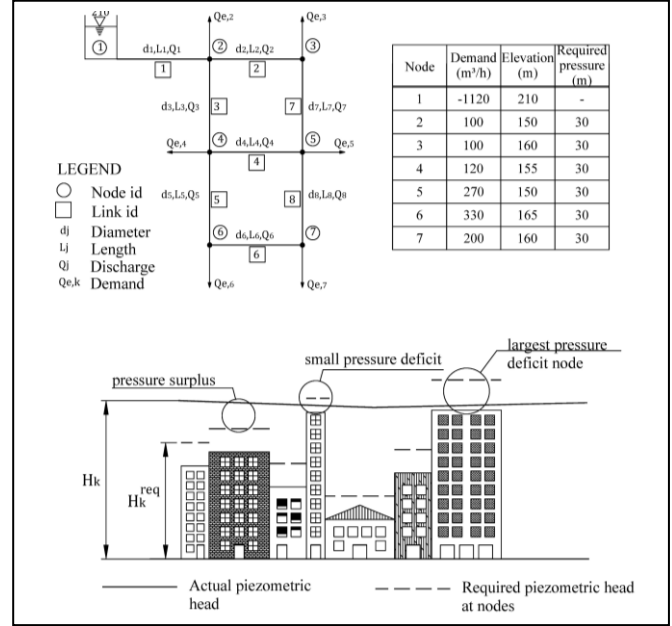
- ΔE_j is the head loss along path j : in a closed loop $\Delta E_j = 0$

The optimization problem is solved through a coupling between multi-objective GAs and the EPANET v. 3.0 hydraulic network simulator [22], adopted to solve the system of equations (3) and (4). In particular, the following expression for the unit head loss in a pipe has been assumed (C_{HW} is the Hazen-Williams roughness coefficient):

$$h_f = 10.668 \frac{Q^{1.852} L}{C_{HW}^{1.852} d^{4.871}} \quad (7)$$

B. Benchmark network adopted for the analysis

The two-loop network shown in Fig. 1 was selected for the numerical simulations: it is a well-known literature system adopted for benchmark, [1]; all the pipelines have a length of 1000 m each, and a Hazen-Williams coefficient equal to 130. Fig. 1 (above) shows the nodal characteristics, while the available commercial diameters are (in inches): 1, 2, 3, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, with unit costs (in \$/m), respectively, of 2, 5,



8, 11, 16, 23, 32, 50, 60, 90, 130, 170, 300, 550. Fig. 1 (below) shows a sketch with the definitions of the actual and the required piezometric heads at nodes, together with some examples of pressure deficit or surplus.

Fig. 1. Above: Two-loop network selected for the simulations, and nodal data. Below: schematic representation of actual and required piezometric head at nodes.

C. Complete enumeration of the search space and Pareto optimal solutions (POS) belonging to the frontier

The search space of the optimization problem is made up of 14^8 solutions and has been entirely explored with an enumerative technique. In other words, all solutions have been simulated and their main characteristics have been calculated. This allowed the derivation of the true Pareto frontier of the problem, which has been limited to configurations ranging from the one with the lowest cost satisfying all pressure constraints, 419,000 \$, to that with a maximum cost of 500,000 \$. This region has been called the Region Of Interest (ROI), and is useful for the calculation of some performance metrics, as explained thereafter.

In particular, 11957 are the solutions within such region of the objective function space, and only 33 of them are Pareto optimal: Tab. I lists their characteristics in terms of the value of the two objective functions, while Fig. 2 their representation in the ROI. Notable is the great difference in pressure surplus

between the first and the second POS, indicating an increase in redundancy with little additional cost.

TABLE I. PARETO OPTIMAL SOLUTIONS OF THE PROBLEM

N.	Cost (\$)	Average surplus (m)	N.	Cost (\$)	Average surplus (m)
1	419000	6.9956	18	460000	12.5351
2	420000	9.8255	19	463000	12.5485
3	423000	9.8381	20	466000	12.9696
4	427000	9.8808	21	469000	12.9859
5	430000	9.9017	22	472000	13.0174
6	436000	10.7175	23	475000	13.0586
7	438000	10.8633	24	476000	13.4298
8	439000	11.4440	25	479000	13.4560
9	442000	11.5177	26	482000	13.5157
10	445000	11.6188	27	485000	13.6001
11	446000	11.8679	28	488000	13.6664
12	448000	11.9876	29	490000	13.7791
13	451000	12.0117	30	493000	13.7834
14	452000	12.0302	31	494000	13.8983
15	454000	12.0610	32	497000	13.9918
16	455000	12.1849	33	500000	13.9999
17	458000	12.4410			

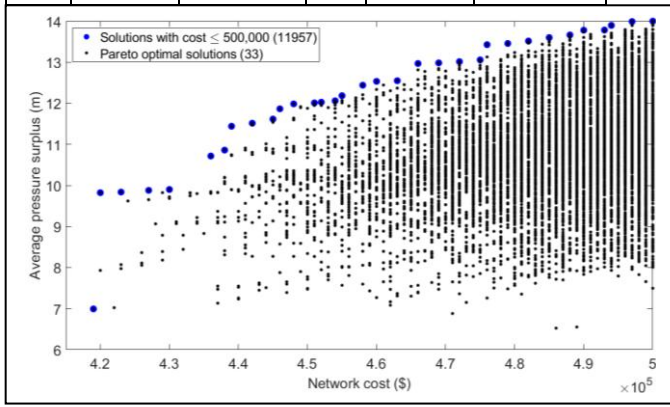


Fig. 2. Representation of all the solutions within the region of interest, with the Pareto optimal solutions evidentiated

III. MULTI-OBJECTIVE GAS AND PERFORMANCE METRICS

A. NSGA-II and Controlled NSGA-II (CNSGA-II)

NSGA-II is an algorithm originally introduced by [13]. Its main advantage is that it relies on a crowded tournament selector based on the rank of the competing individuals and on their crowding distance. Thus, NSGA-II incorporates elitism and it does not need to set the value of the niche radius, which is a parameter that typically affected the performance of the previously introduced NSGA.

However, NSGA-II suffers because of the way elitism is performed: in multi-objective optimization, all solutions residing in the first rank are equally good, and have to be considered as elite individuals to be kept from one generation to the next. But, during the evolution of NSGA-II, often the current population is mostly made of individuals of the first rank, and these might still be far from the true Pareto front. In such case, the elitist approach causes a removal of solutions of non-elitist fronts and, as a result, the search process may suffer from stagnation or premature convergence.

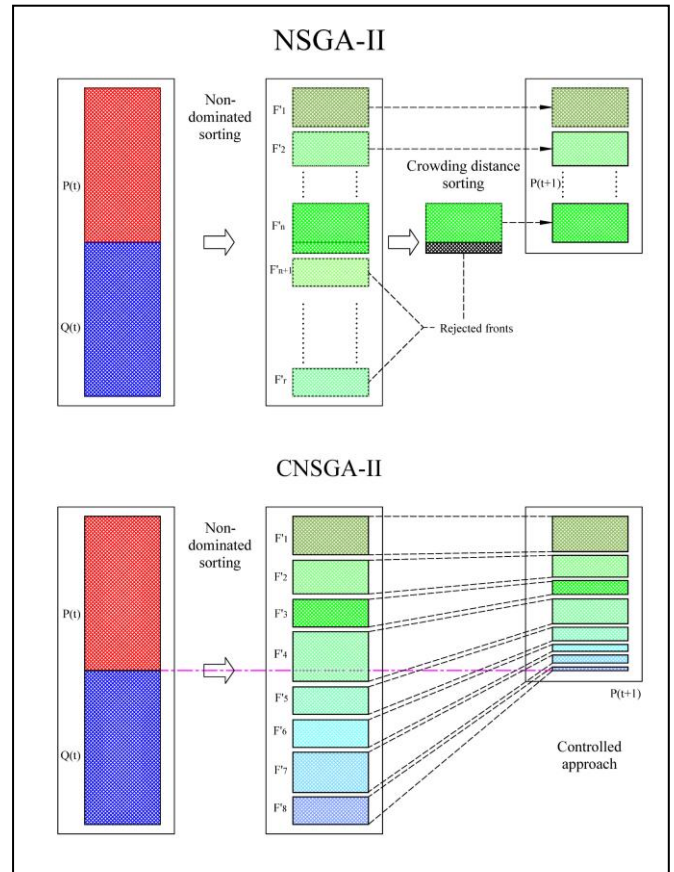
The Controlled NSGA-II is an algorithm developed by [23]: it is very similar to NSGA-II, but it performs elitism in a controlled way, that is, instead of keeping all individuals of rank one, each front is forced to have an exponentially decreasing number of solutions. In this way, a subset of all non-dominated fronts are forced to coexist in the new population, which is typically more diverse than that of NSGA-II. Fig. 3 shows the difference between the algorithms.

If we denote with N the size of the population and with K the number of fronts into which the current population has been sorted, the number of individuals belonging to the i -th front, n_i , admitted to the next population is given by the following relation (r is a reduction factor, $r < 1$):

$$n_i = N \frac{1-r}{1-r^K} r^{i-1} \quad (8)$$

B. Improvements for the algorithms

In analyzing the problem of WDS optimization, there are some issues that should be addressed and that the algorithms



proposed in the literature do not directly take into account. A first consideration derives from the huge extension of the Pareto frontier: in fact, there are solutions that, although mathematically optimal, are of no practical interest; therefore, the problem of limiting the search to a small area of the frontier arises.

Fig. 3. Schematic representation of elitism in NSGA-II and Contolled NSGA-II.

A second issue concerns the discrete nature of the problem, so that the search space is actually formed by a finite set of possible configurations. During the evolution process, the presence of several copies of the same solutions in the optimal front may be a cause of premature convergence, preventing the exploitation of the algorithm.

The first issue has been addressed by adding to the second objective function a negative penalty term which depends on the deviation from the maximum allowable budget, B ($p > 0$ is a penalty factor):

$$f_2^* = f_2 - p \cdot \left[\sum_{i \in D} \sum_{j=1}^{N_p} c(d_i) L_{ij} - B \right] \quad (9)$$

In other words, we penalize the redundancy of those solutions having cost beyond that of the budget (for the problem examined, $B = 500,000$ \$).

The second problem was solved by introducing an additional dominance, the *epsilon* dominance. For a problem with two objectives, the first to be minimized and the second to be maximized, the classical Pareto dominance can be written as:

$$(f_1^a \leq f_1^b \cap f_2^a \geq f_2^b) \cap (f_1^a < f_1^b \cup f_2^a > f_2^b) \quad (10)$$

while in the case of the epsilon dominance, the previous relation is transformed into (ε_1 and ε_2 are small numbers):

$$[(f_1^a \leq f_1^b + \varepsilon_1) \cap (f_2^a \geq f_2^b - \varepsilon_2)] \cap [(f_1^a < f_1^b + \varepsilon_1) \cup (f_2^a > f_2^b - \varepsilon_2)] \quad (11)$$

The two algorithms described above, NSGA-II and CNSGA-II, have been modified in order to implement the concept of epsilon dominance: the aim of the numerical simulations was to verify the impact of these variations on the performance of the algorithms.

C. Performance metrics

The analyses carried out included two types of performance indicators, namely:

- the *hypervolume* (HV), a single scalar quantifying the amount of the objective function space dominated by the Pareto front that the algorithm gradually evolves: since the exact frontier of the problem is known, it was decided to normalize this index with respect to the maximum value of the frontier in the ROI;
- two indices that quantify, respectively, the convergence of the current front towards the real one (*convergence index*, CI, a measure for the exploitation) and the distribution of individuals of the generic population along the current front (*sparsity index* of solutions, SI, a measure for the exploration).

The convergence index, CI, is calculated as:

$$CI = \frac{N_f^{POS}}{N^{POS}} (1 - \bar{\delta}) \quad (12)$$

where N_f^{POS} represents the number of Pareto-optimal solutions (POS) ‘found’ by the algorithm at the generic iteration; $N^{POS} = 33$ is the dimension of P^* , that is, the set of POS reported in Tab. I. $\bar{\delta}$ indicates the average of the Euclidean distances (normalized with respect to the ROI) of all the individuals residing within the ROI from the respective closest Pareto-efficient solution, and is expressed as:

$$\bar{\delta} = \frac{1}{N^{ROI}} \sum_i \delta_i = \frac{1}{N^{ROI}} \sum_i \sqrt{\left(\frac{f_{1,i}^{POS} - f_{1,i}}{f_{1,max} - f_{1,min}} \right)^2 + \left(\frac{f_{2,i}^{POS} - f_{2,i}}{f_{2,max} - f_{2,min}} \right)^2} \quad (13)$$

where N^{ROI} is the number of individuals at the generic generation within the region of interest, $f_{1,i}$ and $f_{2,i}$ are, respectively, the cost and the average pressure surplus of the i -th individual, $f_{1,i}^{POS}$ and $f_{2,i}^{POS}$ the same quantities referred to the Pareto-optimal solution closest to the i -th individual. (The fact that there exists at least one individual closest to a POS is a necessary condition to qualify it as being *reached*, and is important for the calculation of the sparsity index.) $f_{1,max} = 500,000$, $f_{1,min} = 419,000$, $f_{2,max} = 13.9999$, $f_{2,min} = 6.9956$; these four values delimiting the ROI are reported in Tab. I.

The sparsity index, SI, is calculated as:

$$SI = \frac{N_r^{POS}}{N^{POS}} \left(1 - \frac{z_{max}}{N^{POS}} \right) (1 - \sigma_{adim}) \quad (14)$$

where N_r^{POS} represents the number of POS in P^* that have been reached by at least one individual in the current population. Of course, every solution found by the algorithm lying on the real frontier is also reached, but the converse is not necessarily true. z_{max} is the maximum number of adjacent optimal solutions not reached, and σ_{adim} takes into account the actual distribution of the individuals around the reached Pareto-optimal solutions. For further details about the indices see [21].

IV. RESULTS AND DISCUSSION

The simulations performed, on one hand, had the aim of comparing the hypervolume metric to the other proposed performance indices; on the other, they had the purpose of evaluating the improvement, if any, related to the introduction of the epsilon dominance.

Tab. II and Tab. III report the results for NSGA-II and CNSGA-II, respectively, obtained with populations of 50 individuals, mutation probability of adjacent type equal to $1/8 = 0.125$, and 500 generations per run. Note that the mutation probability is the inverse of the chromosome’s length, and has been kept constant throughout the simulations. The crossover probability, of uniform type, was varied between 0 and 1. The results represent the average and standard deviation values obtained after carrying out 10 runs for every crossover probability, each with a different initial seed.

TABLE II. RESULTS FOR NSGA-II

Performance metrics	Crossover probability, p_c					Mean
	0.00	0.25	0.50	0.75	1.00	

HV_{mean}	0.924	0.924	0.915	0.947	0.990	0.940
$HV_{std\ dev}$	0.142	0.142	0.138	0.104	0.009	0.107
CI_{mean}	0.639	0.651	0.477	0.521	0.719	0.601
$CI_{std\ dev}$	0.322	0.326	0.362	0.332	0.203	0.309
SI_{mean}	0.684	0.691	0.612	0.636	0.735	0.672
$SI_{std\ dev}$	0.190	0.207	0.205	0.183	0.090	0.175
$N_f^{POS}_{mean}$	21.1	21.5	15.8	17.3	23.8	19.900
$N_f^{POS}_{std\ dev}$	10.6	10.8	11.9	10.9	6.6	10.160
$N_r^{POS}_{mean}$	25.2	25.3	24.5	25.2	28.1	25.660
$N_r^{POS}_{std\ dev}$	4.6	4.9	4.8	4.1	2.0	4.080
$Z_{max, mean}$	3.4	3.3	3.4	2.7	2.0	2.960
$Z_{max, std\ dev}$	2.8	2.9	2.8	2.1	0.6	2.240

TABLE III. RESULTS FOR CNSGA-II

Performance metrics	Crossover probability, p_c					Mean
	0.00	0.25	0.50	0.75	1.00	
HV_{mean}	0.985	0.955	0.951	0.956	0.987	0.967
$HV_{std\ dev}$	0.016	0.106	0.105	0.106	0.011	0.069
CI_{mean}	0.607	0.632	0.595	0.687	0.658	0.636
$CI_{std\ dev}$	0.298	0.270	0.296	0.293	0.248	0.281
SI_{mean}	0.659	0.647	0.610	0.685	0.715	0.663
$SI_{std\ dev}$	0.161	0.158	0.173	0.165	0.104	0.152
$N_f^{POS}_{mean}$	20.1	20.9	19.7	22.7	21.8	21.040
$N_f^{POS}_{std\ dev}$	9.8	8.9	9.7	9.6	8.1	9.220
$N_r^{POS}_{mean}$	26.0	25.4	24.7	26.8	27.7	26.120
$N_r^{POS}_{std\ dev}$	3.0	4.0	4.2	4.3	2.2	3.540
$Z_{max, mean}$	1.8	2.6	2.7	2.5	2.0	2.320
$Z_{max, std\ dev}$	0.4	2.2	2.1	2.2	0.8	1.540

Tables IV and V report the results with reference to the modified algorithms, the parameters remaining unchanged. From the results, some considerations can be drawn.

Although there is a general agreement between the values of the hypervolume and those of the sparsity and convergence indices, there are also some situations in which high values of the hypervolume do not correspond to better values of the others. One reason is because there are some Pareto optimal solutions whose contribution to the hypervolume is only marginal. In addition, such discrepancy depends on the shape of the frontier and, in particular, on the values of the two objective functions of the POS: the problem examined is discrete in nature, and, given a value of the cost (first objective), several solutions have that value, with different diameters, but with tiny differences in the pressure surplus (second objective). This gives rise to very similar hypervolume indices, which is a geometric measure in the objective space, but likely different values of the other metrics, in particular the convergence index and the number of POS found, N_f^{POS} . On the contrary, the hypervolume is well

correlated with the number of POS reached, N_r^{POS} , especially with associated low standard deviation.

The CNSGA-II is better than the NSGA-II with respect to all performance indices and is also characterized by lower standard deviation values.

There is a notable dependence of the performance of the algorithms on the variation of the crossover probability. On the contrary, the mutation probability was kept constant, because some simulations conducted by varying such parameter did not show a similar trend.

Better performances are obtained with the modifications made by adding the epsilon dominance: observing Tab. IV and Tab. V, a clear improvement in the index values can be noticed, especially in the case of the CNSGA-II.

All the algorithms have difficulties in finding the entire Pareto front in one run: this is evident from the values obtained for the N_f^{POS} index, both in terms of the mean and of the standard deviation. As mentioned before, this can be ascribed to the discrete nature of the problem and, above all, to the specific values of the objective functions.

TABLE IV. RESULTS FOR ϵ -NSGA-II

Performance metrics	Crossover probability, p_c					Mean
	0.00	0.25	0.50	0.75	1.00	
HV_{mean}	0.959	0.926	0.959	0.952	0.990	0.957
$HV_{std\ dev}$	0.107	0.143	0.106	0.105	0.008	0.094
CI_{mean}	0.738	0.696	0.760	0.580	0.722	0.699
$CI_{std\ dev}$	0.271	0.353	0.258	0.319	0.189	0.278
SI_{mean}	0.783	0.754	0.775	0.712	0.790	0.763
$SI_{std\ dev}$	0.173	0.232	0.168	0.188	0.108	0.174
$N_f^{POS}_{mean}$	24.4	23.0	25.1	19.2	23.9	23.120
$N_f^{POS}_{std\ dev}$	8.9	11.6	8.5	10.5	6.2	9.140
$N_r^{POS}_{mean}$	28.1	27.5	28.1	26.6	28.3	27.720
$N_r^{POS}_{std\ dev}$	4.3	5.8	4.2	4.3	2.1	4.140
$Z_{max, mean}$	2.4	3.2	2.7	2.6	1.8	2.540
$Z_{max, std\ dev}$	2.2	2.9	2.2	2.3	0.7	2.060

TABLE V. RESULTS FOR ϵ -CNSGA-II

Performance metrics	Crossover probability, p_c					Mean
	0.00	0.25	0.50	0.75	1.00	
HV_{mean}	0.997	0.986	0.991	0.986	0.991	0.990
$HV_{std\ dev}$	0.003	0.013	0.009	0.013	0.009	0.009
CI_{mean}	0.866	0.637	0.768	0.637	0.768	0.735
$CI_{std\ dev}$	0.053	0.272	0.184	0.266	0.165	0.188
SI_{mean}	0.829	0.739	0.781	0.723	0.789	0.772
$SI_{std\ dev}$	0.061	0.120	0.089	0.103	0.104	0.095
$N_f^{POS}_{mean}$	28.6	21.1	25.4	21.1	25.4	24.320

$N_r^{POS_{std\ dev}}$	1.7	8.9	6.0	8.7	5.4	6.140
$N_r^{POS_{mean}}$	30.1	27.9	28.7	27.6	28.7	28.600
$N_r^{POS_{std\ dev}}$	1.4	2.5	1.8	2.0	2.2	1.980
$Z_{max, mean}$	2.0	1.9	2.0	2.1	1.7	1.940
$Z_{max, std\ dev}$	0.8	0.5	0.8	0.7	0.5	0.660

V. CONCLUDING REMARKS

The work has presented the results of a series of simulations aimed at defining the performance of two evolutionary algorithms adopted for the multi-objective, optimal design of pressurized water distribution systems. The two objective functions considered are the total cost and the average pressure surplus, which is a good measure for the redundancy and hence the reliability of the network.

Some ad hoc indices were introduced in order to measure the exploration and exploitation capabilities of the algorithms, and to compare their values to those of the hypervolume, a widely index used in the literature. The algorithms were then modified to overcome problems typically present in the optimization of water distribution networks, that is, limited extension of the Pareto frontier of interest and discrete nature of the problem. The proposed indices allowed to confirm the validity of the hypervolume as a global measure of the performance of a multi-objective GA even in this particular problem case considered, although with some exceptions due to the particular values of the objective functions.

The modifications introduced for keeping into account ϵ -dominance allowed to obtain better results in terms of convergence and sparsity indices, if compared to the original versions of the same algorithms.

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